

Hydrogeology, Geochemistry, and Microbiology Module Report

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In fulfillment of EVS4010 – EVS Field Course

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University of Ottawa

1) Introduction

1.1. Site Description

The site of interest for this report is NCC Property Asset No. 97390 in Ottawa, ON, which currently consists of a 204.7 hectares vacant lot of land owned by the National Capital Commission (NCC) (Amec Foster Wheeler, 2016). This site is informally known as Pine Grove Forest and can be accessed by an NCC National Capital Greenbelt trail head and parking lot (P19) (Amec Foster Wheeler, 2017). It is located along the north side of Leitrim Road between Hawthorne Road and Ramsayville Road (SNC-Lavalin Environment, 2011).

1.2. History of Land Use

This land was formerly leased by the Ontario Ministry of Natural Resources in 1961 for use as a maintenance and storage depot, leading to the development of several buildings, sheds, and an underground storage tank (UST). Once vacated in the early 1980s, all buildings were demolished. However, upon inspection of the site by ADAMAS Engineering Ltd. in 1992, it was found that the UST had contaminated the surrounding soils and groundwater with petroleum products, specifically trichloroethylene (TCE). Further environmental assessments were performed by Aqua Terre from 2000 to 2011 to determine the extent of TCE contamination in groundwater, which consisted of borehole drilling and monitoring well installation, soil sampling, and drivepoint and groundwater sampling. These monitoring wells remain intact to this day and will be used for groundwater sampling and testing in this report (SNC-Lavalin Environment, 2011).

1.3. Surficial Geology of the Site

In the context of surficial geology, the site consists of brown to grey, medium-grained sand, ranging from 1.1 to 4.3 m thick, overlying grey, silty clay. The silty clay layer reaches a depth of approx. 6.7 m below ground surface (bgs), setting the maximum drilling depth completed at the site to date. It is also reported that the bedrock underlying the site extends from approx. 25 to 50 m bgs and is identified as interbedded shale of the Carlsbad Formation (SNC-Lavalin Environment, 2011).

1.4. Objectives of Study

In this study, we aim to conduct additional research on the hydrogeological, geochemical, and microbiological properties of the groundwater at NCC Property Asset No. 97390 to further evaluate its state and TCE contamination status. This study will consist of two major sections:

hydrogeological analysis, and geochemical and microbiological analysis; each comprising of several sub-objectives.

Our hydrogeological research objectives include the construction of a hydrostratigraphic cross section along a line joining monitoring wells, MW15-59, MW10-12, MW10-11, MW15-51, MW10-10, and MW15-50, the mapping of hydraulic head across the shallow aquifer, and the determination of hydraulic conductivity and groundwater flux (specific discharge) across several points.

Our geochemical and microbiological research objectives include the evaluation and plotting of characteristics of groundwater at 9 points inside and outside of the TCE plume, and the determination of main differences in geochemistry and microbiology inside and outside of the plume.

2) Methods

2.1. Hydrogeological Evaluation

On August 25th, 2025, we focused on the hydrogeological components of groundwater, which included the locating of all piezometers (43?), the assessment of water table height and total well depth at each piezometer, and the conduction of slug tests for a series of 5 joined wells.

To locate each well, we used Google Earth to load the existing file of well coordinates for the area and referenced a piezometer location map to ensure correct positioning and identification of wells. Once located, we employed a water level meter (Heron Instruments, “Skinny Dipper”) to determine the depth of the water table (m) and the total depth of the well (m). The depth of the water table was measured from the top of the pipe to the presence of water, and the total depth from the top of the pipe to the top of the well screen at the bottom of the piezometer. We also measured the diameter of the pipe at each well to determine well radius. Through these measurements, we determined the water depth in each well, water volume, hydraulic conductivity, and groundwater flux (specific discharge).

2.2 Geochemical and Microbiological Evaluation

We took water samples from 9 piezometers on site. We began by purging the top layers of water in the piezometers to ensure that oxygenated water was not collected. We calculated the amount of water being purged by determining the volume of water in the piezometers and multiplying by 3. Once purged, the water was extracted from the sample tubes with a syringe using a 3-way valve. 5 syringes were filled per piezometers.

In the field, one syringe was used to measure the redox potential and pH. Measurements were taken every 30 seconds in a 5-minute timeframe as the sample was being added. The alkalinity was also measured in the field. One syringe was filtered using a 0.45µm filter. This

filtered sand, microbes, and precipitates. A Hach digital titrator was used to measure alkalinity with 1.6N (0.8M) sulfuric acid. We measured the amount of acid added for the solution to turn gray, then purple, and finally pink.

In the lab, ion contents were measured from the water samples. Fe³⁺ iron, Fe²⁺ iron, total dissolved iron, total iron, and sulphate. This was measured with the Hach machine, measuring colorimetry. 25 mL of the water sample was added to a test tube with Hach reagents before measuring 10 mL of the solution. Sulphate was the exception, with only 10mL being mixed with the Hach reagent. All the test tubes had the water added through the 0.45µm filter, besides the total iron test. Some of the samples needed to be diluted with DI water for total dissolved iron and for total iron.

3) Results

3.1. Hydrostratigraphy

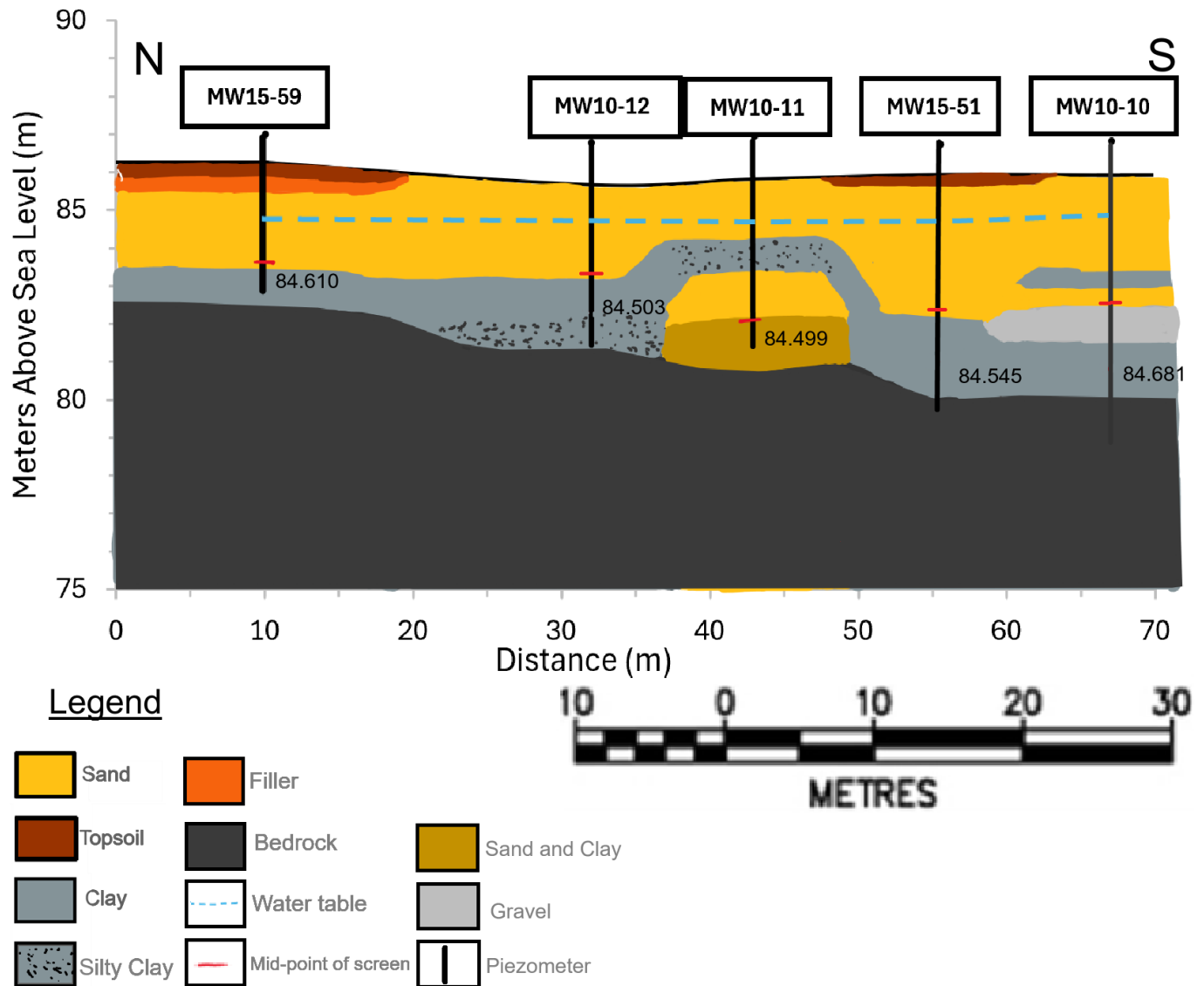


Figure 1. A hydrostratigraphic cross sectional representation of a portion of the wells at the NCC Property Asset No. 97390 area. This includes the geological profile and water table in depth (m). The cross-sectional view depicts well's MW15-59, MW10-12, MW10-11, MW15-51, and MW10-10. The water table is represented and calculated by the hydraulic head.

3.2. Mapping of Hydraulic Head

A total of 5 wells were subject to slug testing at NCC Property Asset No. 97390 on August 25th, 2025: MW10-10, MW10-11, MW10-12, MW15-51, and MW15-59. For all mentioned wells, the well screen length (L) is 1.52 m, and the well screens are intersected by sand. Each well has their respective well casing radius and water depth, shown in Table 1. The Hvorslev method is applicable for this aquifer as the top of the well screen is below the water table and the screen does not fully penetrate the aquifer. The hydraulic conductivity, based on the Hvorslev method and, and groundwater flux (specific discharge) for each well is displayed in Table 1.

Based on the figures below (Figure 3 – 7), we determined that all 5 wells have the presence of groundwater flow and acceptable hydraulic conductivity values when compared to the literature (i.e. clean sand K range is 10^{-5} to 10^{-3} m/s) (Freeze & Cherry, 1979).

Table 1. Hydraulic characteristics of wells used for slug testing. Water level values and casing radiuses were measured in the field on Aug 25th, 2025.

Piezometer #	Water level (m)	Radius of Casing (m)	K (m/s)	q (m ³ /m ² /s)
MW10-10	0.668	0.0155	4.91028E-05	-2.64E-07
MW10-11	0.986	0.016	9.17792E-05	-2.467E-07
MW10-12	1.289	0.0155	0.000187442	-5.178E-07
MW15-51	2.194	0.02	0.000166966	-4.488E-07
MW15-59	0.727	0.02	1.87635E-05	-6.255E-08

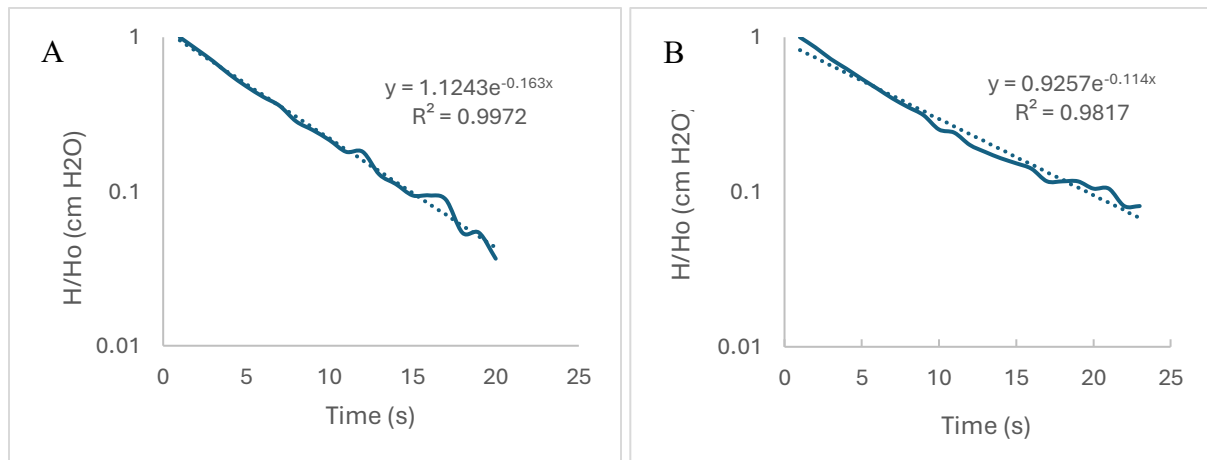


Figure 3. Slug test performed at MW10-10 on Aug. 25th, 2025, at 12:18 PM. H/H_o is represented in logarithmic form to achieve a semi-logarithmic plot. **(A)** Slug was dropped into well and recovers to static H . **(B)** Slug was removed from well and recovers to static H . Both graphs contain a linear-fit equation in exponential form.

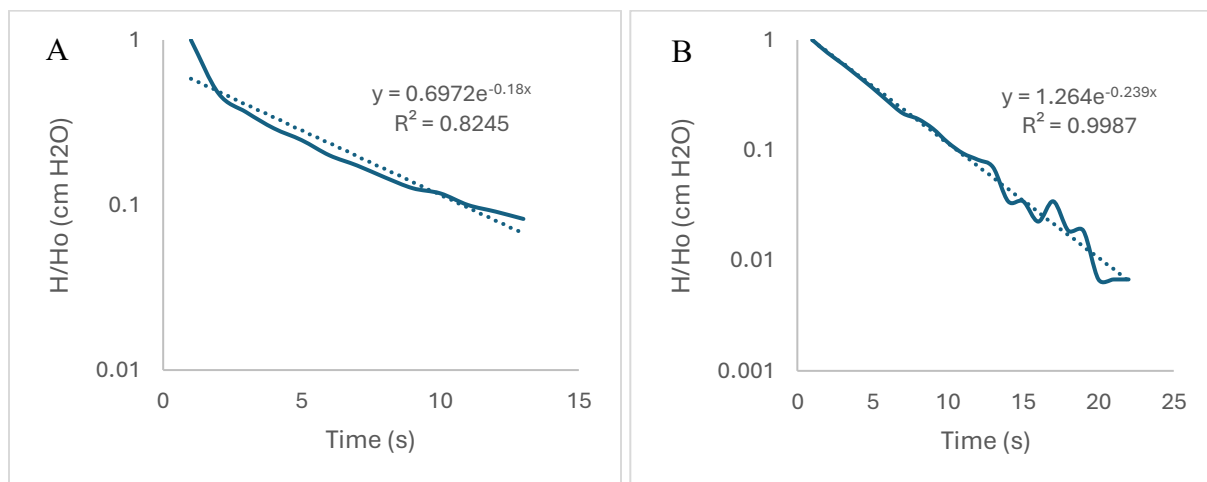


Figure 4. Slug test performed at MW10-11 on Aug 25th, 2025, at 11:30 AM. H/H_o is represented in logarithmic form to achieve a semi-logarithmic plot. **(A)** Slug was dropped into well and recovers to static H. **(B)** Slug was removed from well and recovers to static H. Both graphs contain a linear-fit equation in exponential form.

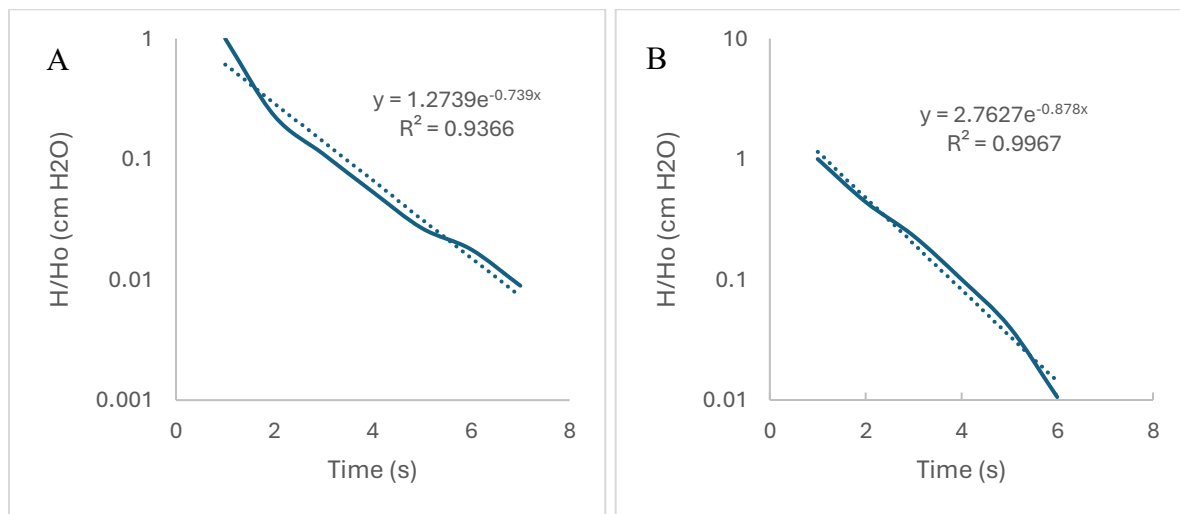


Figure 5. Slug test performed at MW10-12 on Aug 25th, 2025, at 2:25 PM. The duration of the slug drop is represented by the blue line, with the linear portion represented by the orange line. The linear equation, in exponential form, is calculated from the linear portion. H/H_o is represented in logarithmic form to achieve a semi-logarithmic plot.

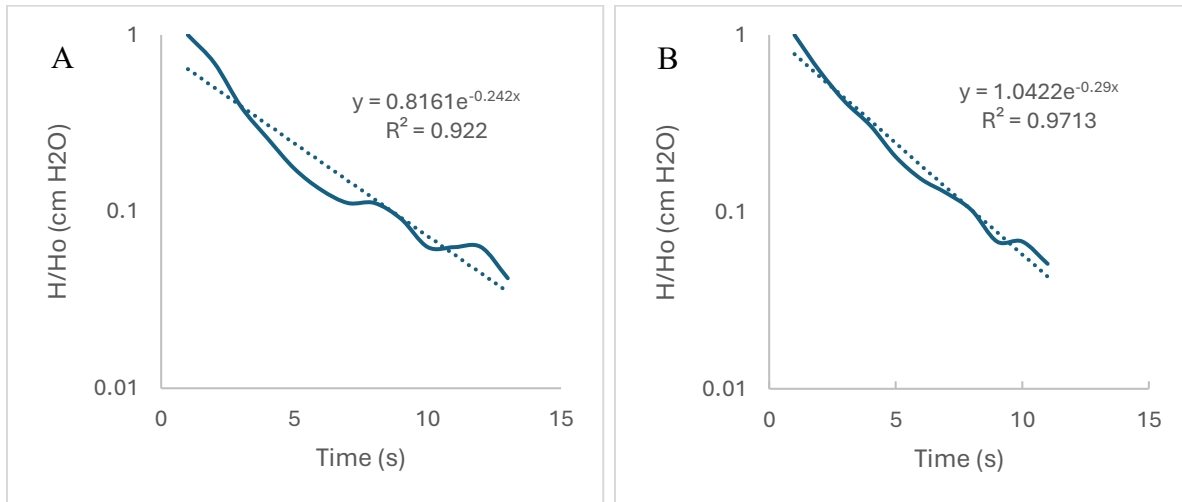


Figure 6. Slug test performed at MW15-51 on Aug. 25th, 2025, at 1:22 PM. The duration of the slug drop is represented by the blue line, with the linear portion represented by the orange line. The linear equation, in exponential form, is calculated from the linear portion. H/H_o is represented in logarithmic form to achieve a semi-logarithmic plot.

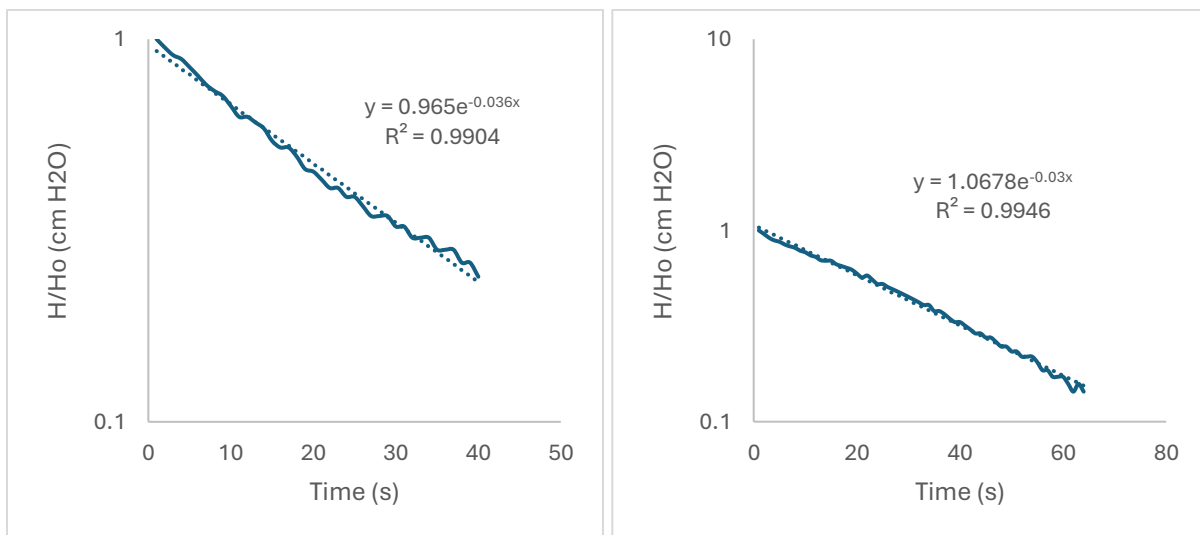


Figure 7. Slug test performed at MW15-59 on Aug 25th, 2025, at 3:09 PM. The duration of the slug drop is represented by the blue line, with the linear portion represented by the orange line. The linear equation, in exponential form, is calculated from the linear portion. H/H_o is represented in logarithmic form to achieve a semi-logarithmic plot.

3.4 Geochemical and Microbiological Parameters

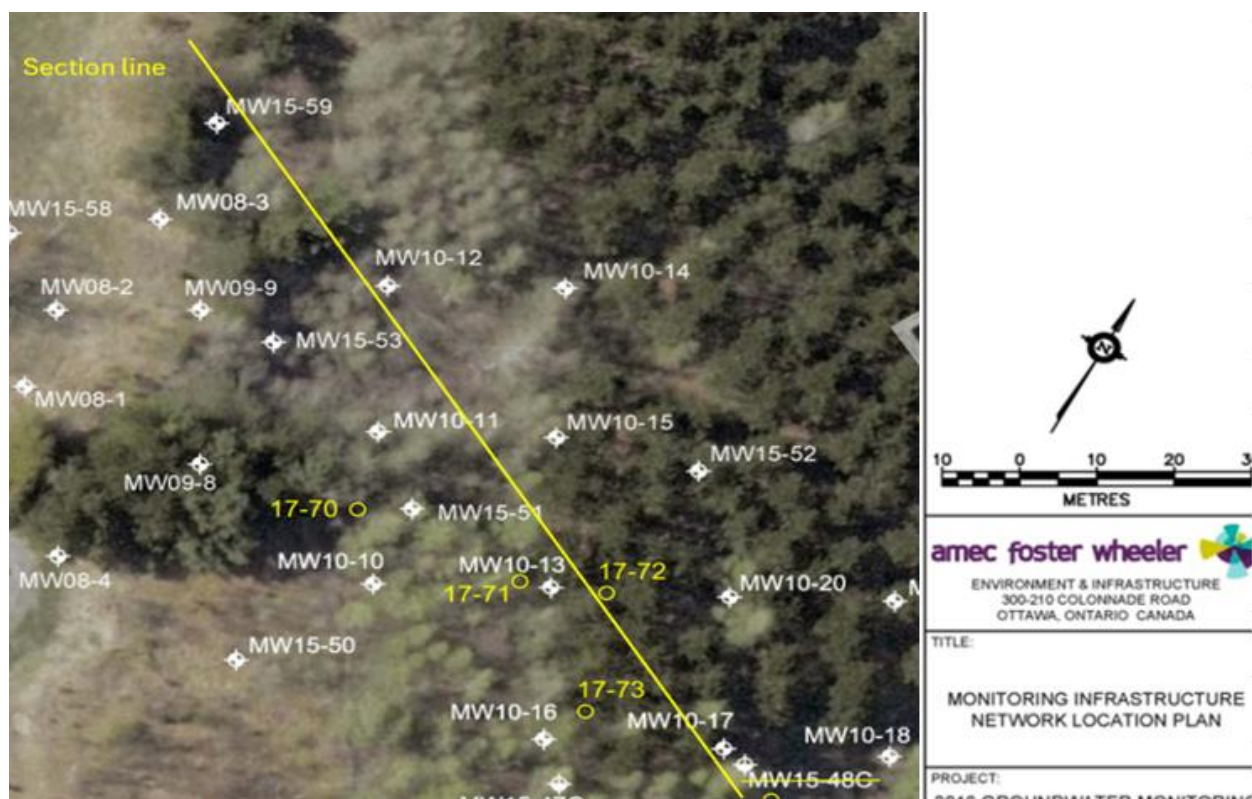


Figure 8. Map of NCC Property Asset No. 97390 showing the section line originating near the well MW15-59. The line passes through wells located near the TCE plume (MW15-59, MW10-12, MW10-14, MW10-11, MW17-70, MW15-51, MW10-10, MW15-52, MW10-13)

Table 2. Geochemical and microbiological characteristics of groundwater at points along the section line(Figure 6), including Distance along the section line(m), PH, redox potential (mv), Alkalinity (CaCO₃ mg/l), Concentration Fe²⁺(mg/l), Fe³⁺(mg/l), Total Fe(mg/l), and So₄²⁻(mg/l).

Piezometer	distance between wells (m)	Distance along the section line(m)	PH	redox potential (mv)	Alkalinity (CaCO ₃ mg/l)	Fe ²⁺ (mg/l)	Fe ³⁺ (mg/l)	Total Fe(mg/l)	So ₄ ²⁻ (mg/l)
MW15-59	0	10	6.56	105	92	0.58	0.39	0.94	15
MW10-12	12	22	6.95	134	156	0.53	0.27	0.3	51
MW10-14	8	30	6.84	157	368	0.04	0	0	0
MW10-11	4	34	7.22	81	304	0.03	0.1	0.33	16
MW17-70	7	41	7.18	104	236	0	0.04	0	31
MW15-51	2	43	7.32	-80	388	1.18	0.13	4.85	20
MW10-10	6	49	7.22	-36	388	0.9	1.21	2.85	43
MW15-52	4	53	7.44	-22	132	0.12	0.64	0.05	18
MW10-13	2	55	7.23	-37	224	1.02	1.24	2.15	42

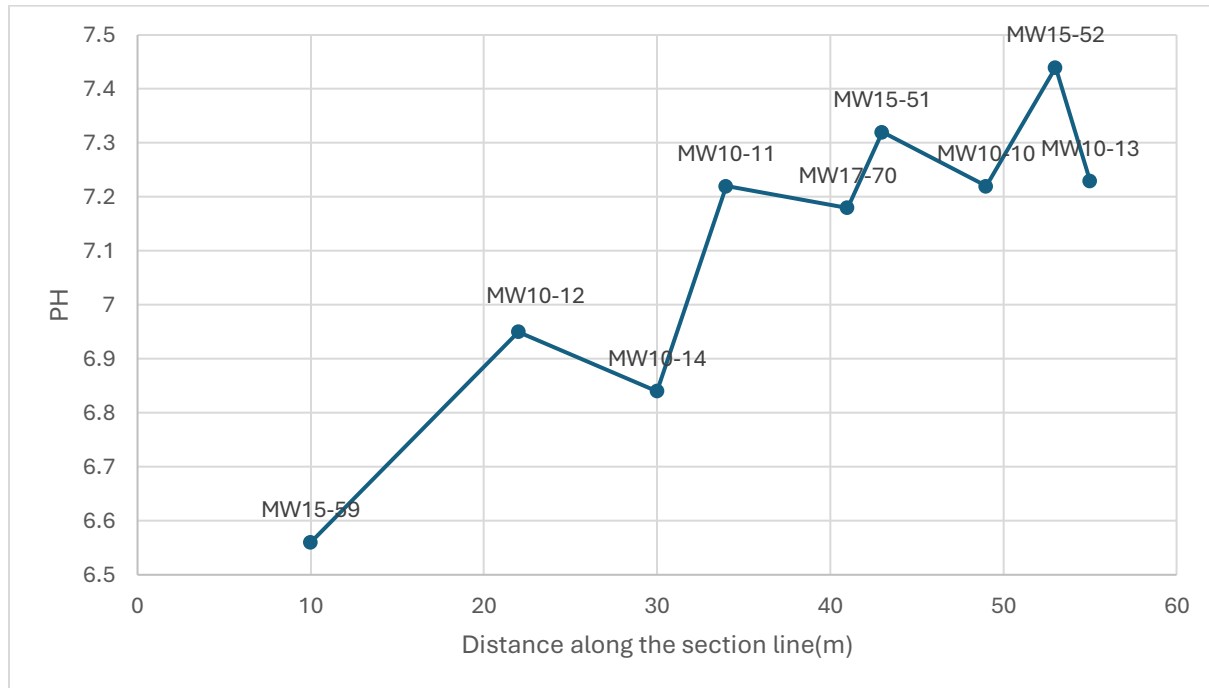


Figure 9. The variation of groundwater PH along the section line with the name of the collected well name.

The PH value is gradually increasing along the section line, the pH was getting higher when it is closer to the east.

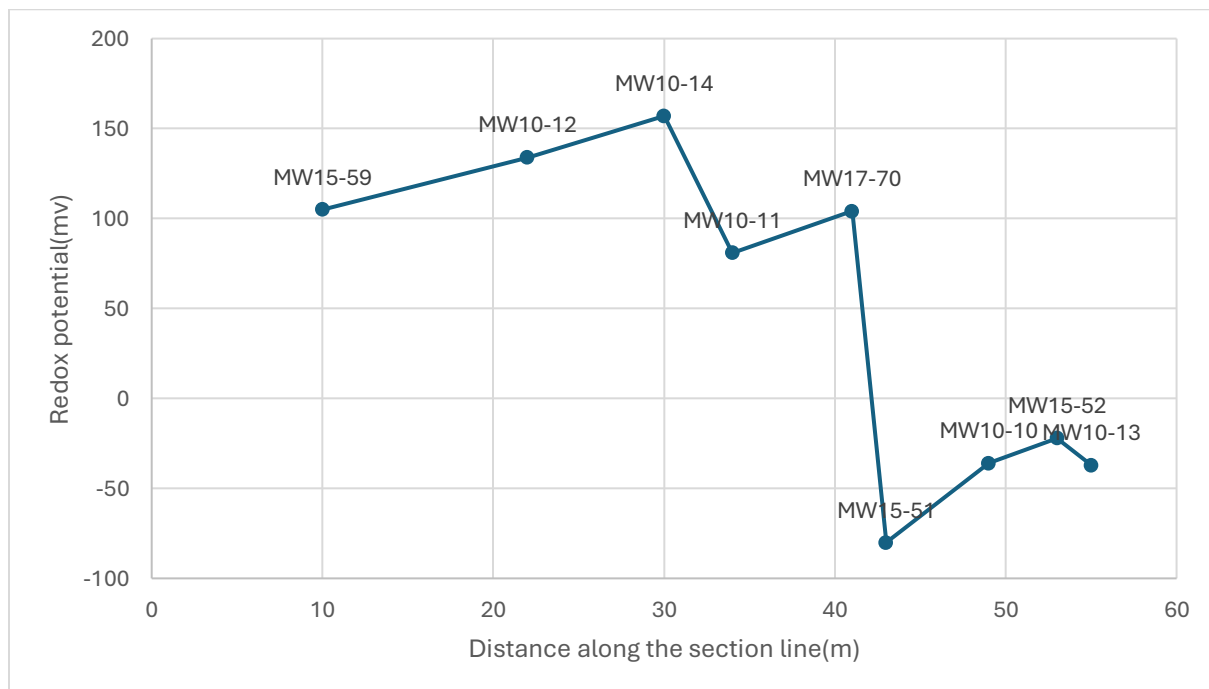


Figure 10. The variation of Redox potential (mv) along the section line with the name of the collected well name.

In Figure 10, the redox potential remains over 50 mv, until 41 m along the section line, then it significantly decreased to -80, and remain negative in the rest distance it became a reducing environment.

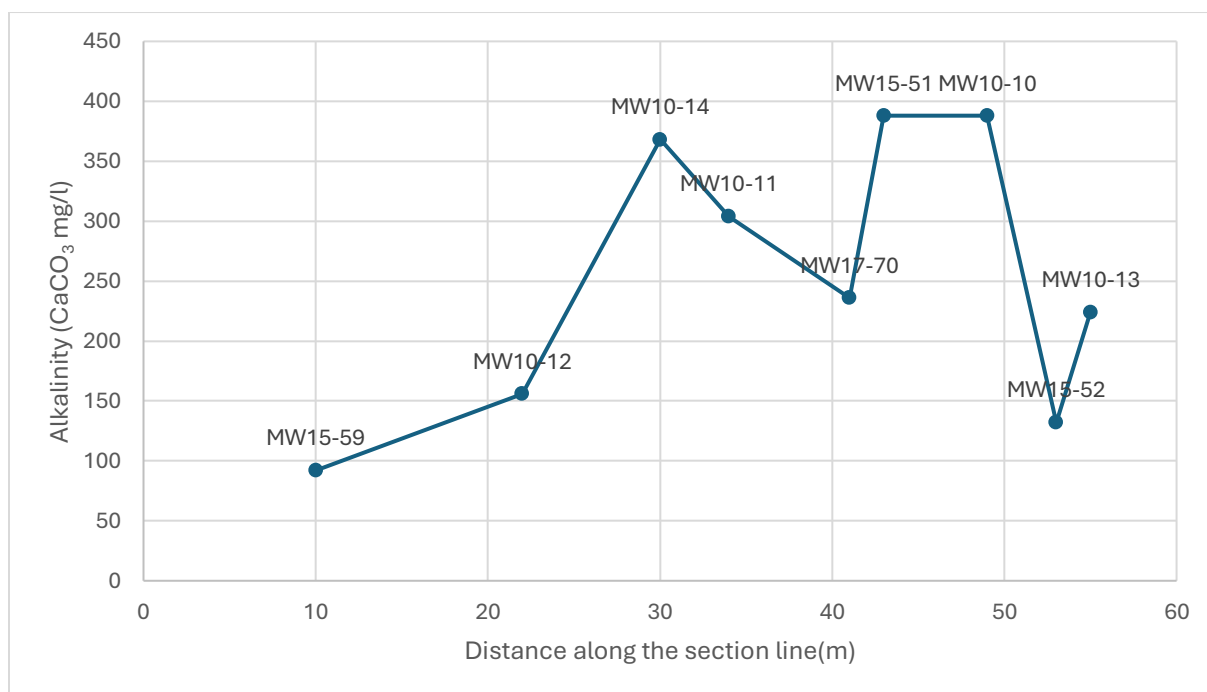


Figure 11. The variation of Alkalinity (CaCO₃ mg/l) along the section line with the name of the collected well name.

In Figure 11, the alkalinity(CaCO₃ mg/l) in MW15-51 and MW 10-10 are similar, the location of the two wells are close.

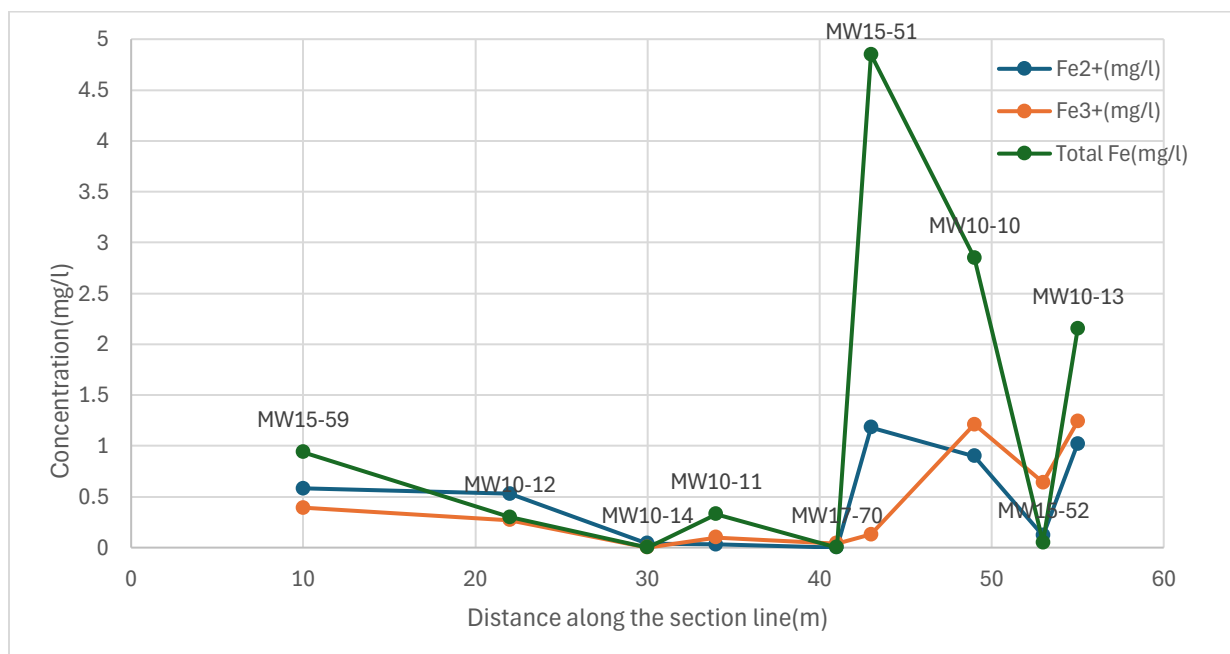


Figure 12. The variation of Concentration (Fe²⁺, Fe³⁺, and Total Fe) (mg/l) along the section line with the name of the collected well name.

In Figure 12, the concentration of total Fe increases significantly at MW15-51 and MW10-10, exceeding 3 mg/L. For most wells, the concentration of Fe^{2+} is higher than that of Fe^{3+} . At MW10-12 and MW15-52, however, the total Fe concentration is lower than either Fe^{2+} or Fe^{3+} , which may have been caused by an experimental error.

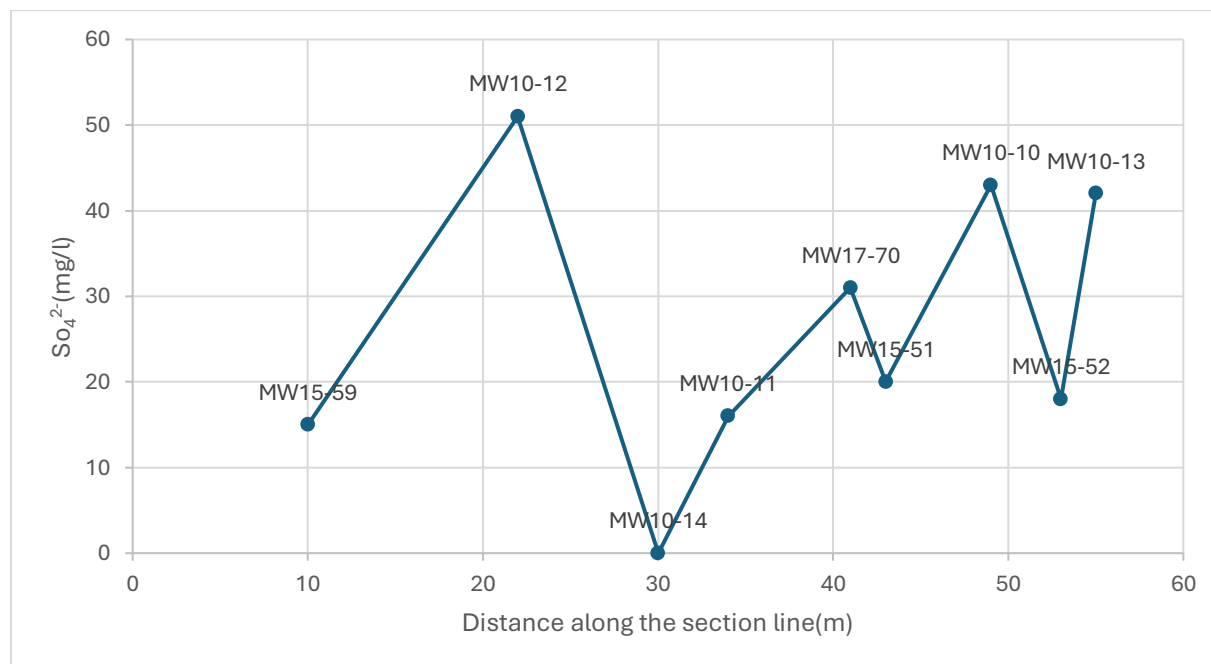


Figure 13. The variation of Concentration of SO_4^{2-} (mg/l) along the section line with the name of the sampled wells.

3.5 Geochemical and microbiological parameters difference between inside (MW17-70, MW10-10, MW10-11, MW10-13, MW15-51,) and outside of the plume (MW10-12, MW10-14, MW15-52, MW15-59).

Table 3. Geochemical and microbiological characteristics of groundwater between inside (MW17-70, MW10-10, MW10-11, MW10-13, MW15-51) and outside of the plume (MW10-12, MW10-14, MW15-52, MW15-59), including PH, redox potential (mv), Alkalinity (CaCO_3 mg/l), Concentration of Fe^{2+} (mg/l), Fe^{3+} (mg/l), Total Fe(mg/l), and SO_4^{2-} (mg/l).

Location	Piezometer	PH	Redox potential (mv)	Alkalinity (CaCO_3 mg/l)	Fe^{2+} (mg/l)	Fe^{3+} (mg/l)	Total Fe(mg/l)	SO_4^{2-} (mg/l)
Inside	MW17-70	7.18	104	236	0	0.04	0	31
	MW10-10	7.22	-36	388	0.9	1.21	2.85	43
	MW10-11	7.22	81	304	0.03	0.1	0.33	16
	MW10-13	7.23	-37	224	1.02	1.24	2.15	42
	MW15-51	7.32	-80	388	1.18	0.13	4.85	20
Outside	MW10-12	6.95	134	156	0.53	0.27	0.3	51
	MW10-14	6.84	157	368	0.04	0	0	0
	MW15-52	7.44	-22	132	0.12	0.64	0.05	18
	MW15-59	6.56	105	92	0.58	0.39	0.94	15

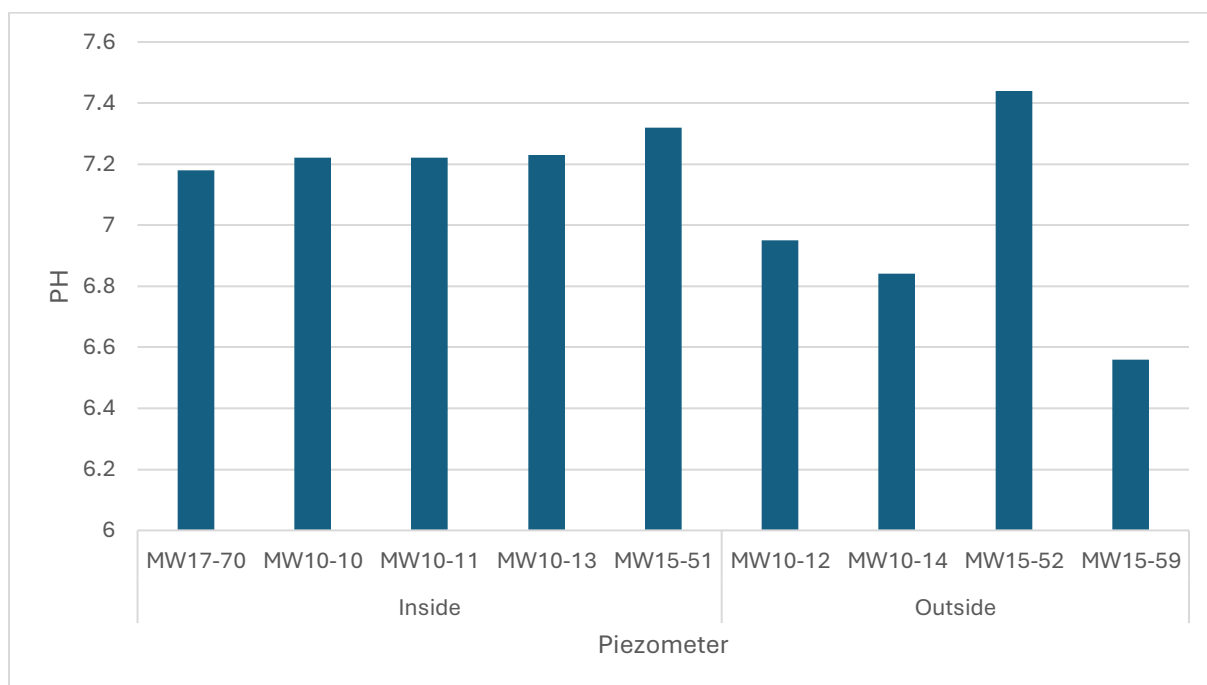


Figure 14. Comparison of groundwater PH values between wells located inside and outside the TCE plume.

In Figure 14, all the wells inside the plume have a pH value higher than 7, whereas 3 out of 4 of the wells outside the plume have a pH value lower than 7.

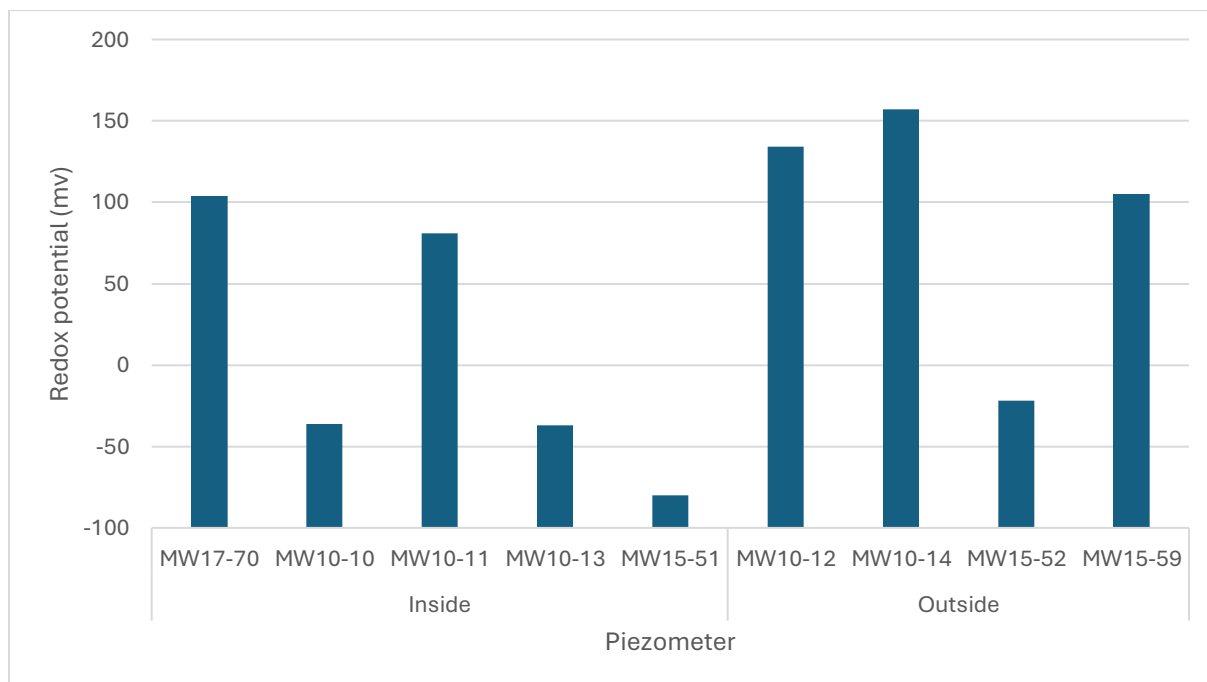


Figure 15. Comparison of groundwater Redox potential (mv) between wells located inside and outside the TCE plume.

The redox potential inside the plume is lower than the redox potential outside the plume. The environment inside the plume is more likely to be reducing.

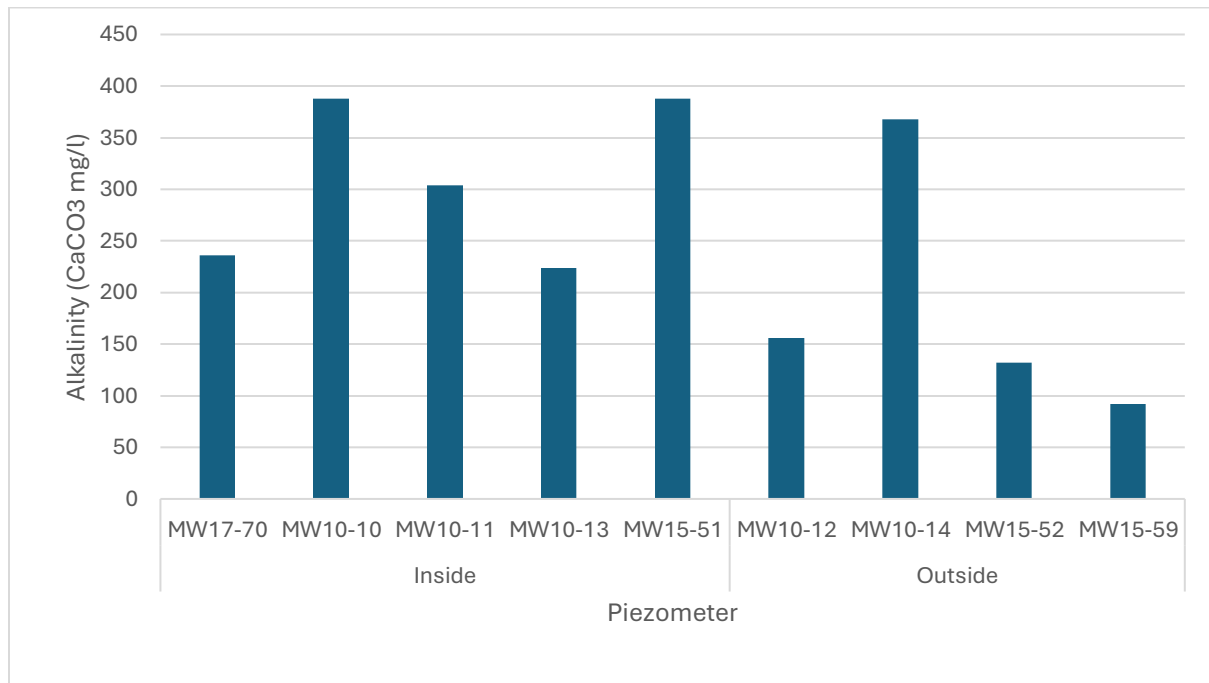


Figure 16. Comparison of groundwater Alkalinity (CaCO₃ mg/l) between wells located inside and outside the TCE plume.

In Figure 16, the alkalinity of groundwater inside the plume is generally higher than the groundwater outside the plume.

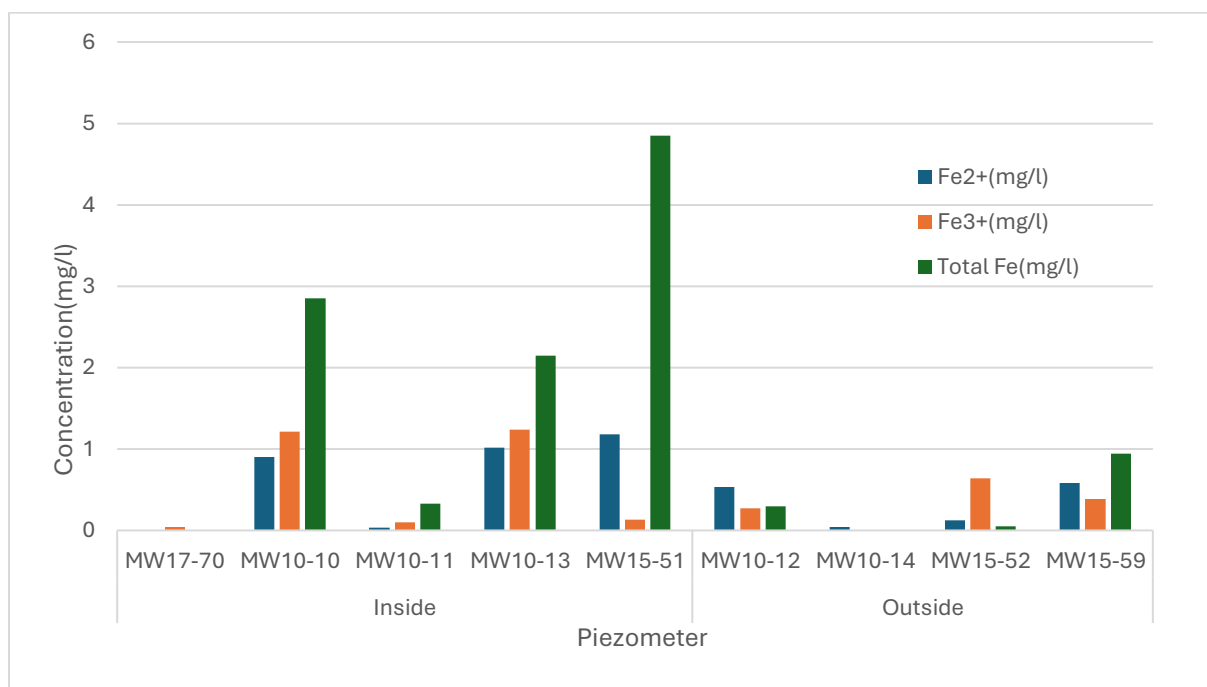


Figure 16. Comparison of groundwater Concentration of Fe^{2+} , Fe^{3+} and total Fe(mg/l) between wells located inside and outside the TCE plume.

In Figure 16, the groundwater inside the plume have concentration of Fe^{2+} , Fe^{3+} and total Fe(mg/l) much higher than the groundwater outside the plume.

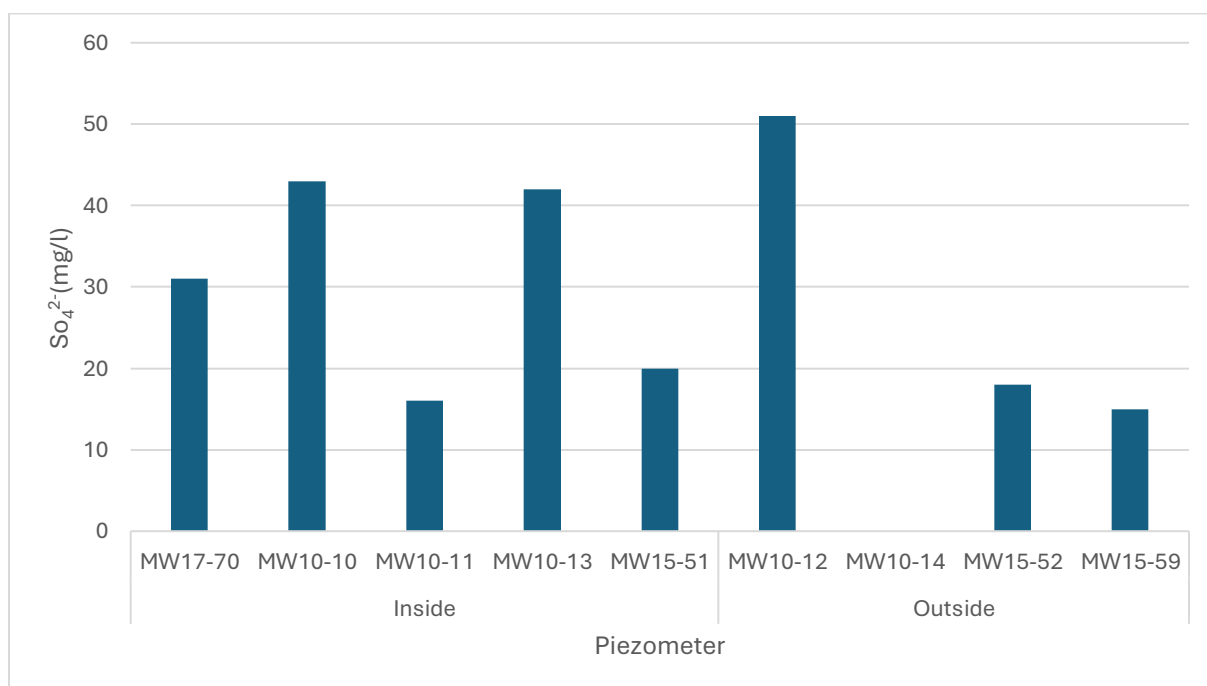


Figure 17. Comparison of groundwater concentration of SO_4^{2-} (mg/l) between wells located inside and outside the TCE plume.

The concentration of SO_4^{2-} (mg/l) inside the plume is slightly higher than outside of the plume.

The geochemical and microbiological characteristics of groundwater reveal that along the section line, the pH value was gradually increasing (Figure 9), all the wells inside the plume have a pH value higher than 7. The redox potential decreased a lot along the section line, the groundwater environment transition from oxidizing to reducing condition. The concentration of Fe in groundwater also increased, particularly at MW15-51 and MW10-10, where total Fe exceeded 3 mg/L, the concentration of Fe inside the plume was significantly higher than outside the plume. For concentration of SO_4^{2-} , it is not stable along the section line, but the concentration inside the plume is higher, most of the Fe and SO_4^{2-} may be contained in the plume. Generally, the groundwater inside the plume is characterized by neutral pH, lower redox potential, higher Fe and SO_4^{2-} concentration compared to the groundwater outside the plume, maintaining a reducing environment, which indicated that the plume retains majority of the contaminants inside.

The groundwater parameters were compared inside the plume versus outside the plume. The pH inside the plume appears to be higher than outside the plume, besides the MW15-52 well, which has a higher pH compared to the rest of the wells. The redox potential appeared to be higher outside of the plume. There was however, one outlier with MW15-52, which had a low redox potential. The alkalinity was higher inside the plume; however, MW10-14 outside the plume had high alkalinity as well. The iron concentration levels were much higher inside the plume compared to outside. The Fe^{3+} values were higher than the Fe^{2+} values inside the plume and Fe^{2+} values were higher, for the most part, outside the plume. Two sample t-tests were performed to determine whether the groundwater parameters between piezometers located inside versus outside the plume were statistically significant. It was found that none of these variables were statistically significant.

4) Discussion

4.1. Hydrostratigraphy

The soil profile identified surrounding the wells were predominantly composed of topsoil, clay, silty clay, and sand. These sediments were deposited during the formation of the Champlain Sea in the Ottawa region approximately 12 Ka (Fulton 1987). The type of aquifer located in the NCC Property Asset No. 97390 area is an unconfined aquifer. There is no impermeable layer existing to prevent run off or precipitation from entering, as the geological composition surrounding the ground water includes clays, silty clays and sand. The understanding that this aquifer is categorized as unconfined is due to the knowledge that precipitation and run off seeps into the aquifer. Additionally, since there has been contamination

of organic compounds entering the groundwater system, which further supports the characterization of the aquifer.

4.2 Slug Test and Hydraulic Conductivity

The slug tests performed were used to determine the hydraulic conductivities and the fluxes of each corresponding well. The slug test allows for the disturbance of equilibrium in the well to produce an instantaneous change in the hydraulic head (the water level) to achieve the unrecovered drawdown. During the slug tests conducted, the highest conductivities observed were at MW10-12 and MW15-51. Hydraulic conductivity values of $2.17 \times 10^{-4} \text{ LT}^{-1}$ and $1.74 \times 10^{-4} \text{ LT}^{-1}$ of the slug going in, and $1.58 \times 10^{-4} \text{ LT}^{-1}$ and $1.67 \times 10^{-4} \text{ LT}^{-1}$ for the slug being removed were yielded. The higher hydraulic conductivities are derived from piezometers that present a similar geologic profile which include similar depths of sand and clay. This indicates that the ease at which the ground water was able to flow through the geologic material is high.

4.3

Based on the data from Figure 2, the water in the aquifer is flowing eastward. Given that the water in the aquifer is flowing eastward, from high hydraulic head to low hydraulic head, it can be inferred that contaminants from the plume arrived from the west, and contaminants could potentially migrate towards the east. When preventing the spread of contaminants, it is important to focus on concentrating remediation efforts on the east of the plume, as well as directly on the plume.

4.4 Geochemical and microbiological characteristics

When ferric iron is present such as in MW10-10 and MW10-13, the water is likely anaerobic, because Fe^{3+} precipitates as iron oxide. Ferrous iron is more soluble and will exist in aerobic conditions, such as in MW15-51. Sulphide is the most reduced form of sulphur. It appears when sulphate reducing bacteria is present. They oxidize organic carbon and transfers electrons to sulphate. Where sulphate was present in the wells, there were likely to be sulphate-reducing bacteria. These bacteria exist in anoxic conditions, meaning the wells were in anoxic conditions. Black precipitate was found at the bottom of sampling tubes in the wells. This precipitate is iron sulphide. It is possible that the sulphate-reducing bacteria were present in oxygenated conditions due to their abilities to go dormant (sporulating) (Fortin, 2025).

The redox potential was lower in the plume because of the reducing power of microbes. A lower redox potential indicates stronger reducing abilities. Sulfur and iron are also redox sensitive. According to the data, when the redox potential was higher (MW15-51, MW10-10, MW10-13), the sulphate presence was high. Alkalinity was higher inside the plume because Fe is reduced. Fe reduction produces bicarbonate species. Bicarbonate and carbonate species increase the buffering capacity. pH was generally lower outside the plume. This is because there are less reducing

reactions present outside of the plume, which produce H^+ . H^+ increases the pH, which explains why the pH is higher in the plume.

Sources of Error

During the visit to the NCC Property Asset No. 97390 site, Ottawa had been currently undergoing a through a period of a lack of precipitation. This caused water levels in wells to significantly drop, disabling certain wells from having tests being conducted on them. Once water samples were collected through syringes, air bubbles trapped within them must be expelled. This was done by pushing on the plunger and tapping on the syringe. However, extremely small air bubbles would remain trapped and were unable to escape from the syringe, which may have contaminated samples that derived from anoxic condition. While visiting each well to conduct water level measurements, the instrumentation used was highly sensitive to indicating water emergence. While this was beneficial to identify water levels, this also created sources of error. Where water levels were improperly measured by being misled by the sensor when it would indicate the presence of water when it would be as simple as condensation. Producing an instantaneous change in hydraulic head left much room for error, making it difficult to conduct especially by hand. This left room for error found in the data of the slug test including any significant oscillations in the data sets.

5) Conclusion

In our evaluation of NCC Property Asset No. 97390, we determined and visualized several hydraulic and geologic properties of the wells, as well as the microbiological and geochemical characteristics of the groundwater.

Through the mapping of hydraulic head across the aquifer, we determined that the groundwater flow is heading in the eastward direction, flowing from high to low elevation. In the context of the surficial geology underlying wells MW15-59, MW10-12, MW10-11, MW15-51, and MW10-10, we observed that the main sediment types include sand, silty sand, and clay, with all well screens adjacent to sand. The presence of sand beside the well screens is in support of our calculated hydraulic conductivities for each well, as all K values ranged from 10^{-4} to 10^{-5} , signifying ideal groundwater flow conditions.

Our geochemical analysis determined that the groundwater inside of the TCE plume is under reducing conditions compared to outside, characterized by a low redox potential and high dissolved Fe and total Fe conditions. These characteristics suggest the presence of sulfate-reducing bacteria (SRBs) in the groundwater inside of the plume.

In general, our investigation of the groundwater NCC Property Asset No. 97390 suggests that remaining contaminants in the TCE plume will be in their reduced state, due to the low redox potential of the environment as a result of SRB activity, and may require further evaluation into the potential environmental risks of these compounds.

References

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Appendices

piezometer_# (W to E)	depth_to_water	total_depth	water_level/pressure_head
MW15-62	2.624	3.669	1.045
MW15-56	2.982	3.537	0.555
MW15-55	2.893	3.077	0.184
MW08-5	1.793	3.137	1.344
MW15-57	2.948	4.592	1.644
MW15-54			
MW13-27	1.944	4.267	2.323
MW15-58	2.599	3.155	0.556
MW08-1	2.555	4.958	2.403
MW08-02	2.642	5.225	2.583
MW08-4	2.382	5.042	2.660
MW13-29	1.958	5.614	3.656
MW08-3	2.370	5.150	2.780
MW09-9	2.326	3.648	1.322

MW09-8			
MW15-59	2.400	3.127	0.727
MW15-50			
MW15-53	1.908	2.614	0.706
MW10-10	2.469	3.137	0.668
MW10-11	2.461	3.447	0.986
MW10-12	2.277	3.566	1.289
MW15-51	2.235	4.429	2.194
MW15-45	2.626	5.209	2.583
MW10-16	2.754	4.227	1.473
MW10-13	2.918	3.549	0.631
MW10-15	2.949	3.536	0.587
MW10-14	2.998	3.287	0.289
MW10-24	2.238	3.594	1.356
MW15-52	2.681	3.777	1.096
MW15-39	3.079	4.022	0.943
MW10-17	3.146	5.057	1.911
MW10-20	0.000	2.855	
MW15-41	2.680	3.539	0.859
MW15-40C	2.634	7.592	4.958
MW10-25	2.962	3.904	0.942
MW10-18	2.899	4.173	1.274
MW10-19	2.925	4.083	1.158
MW15-63	2.445	3.602	1.157
MW15-42C	2.915	6.972	4.057
MW15-43	2.428	2.826	0.398
MW10-21	2.432	3.524	1.092
MW10-23	2.241	3.535	1.294
MW10-22	2.066	3.555	1.489

A-1. Water level data of piezometers

Subsurface Profiles			
	soil type	to elevation (m)	to depth (m)
15-51	topsoil	85.93	0

	sand	82.28	3.66
	clay	79.84	6.1
15-59	topsoil	86.26	0
	fill	85.5	0.46
	sand	83.97	2.29
	clay	83.21	3.05
MW10-10	sand + trace gravel	85.918	3.08
	clay	82.838	0.21
	sand	82.629	1
	gravel	81.629	1.2
	clay	80.429	1.2
MW10-11	sand	85.816	2.25
	silty clay	83.566	0.77
	sand	82.796	
	sand and clay		
MW10-12	sand	85.662	2.57
	clay	83.092	0.5
	silty clay	82.592	1.2

A-2. Subsurface profiles

piezo mete rs	easting	northin g	ground elev (masl)	borehole depth (m)	borehole bottom elevation (masl)	top of PVC elev atio n	well scre en leng th (m)	Geol ogic Medi a Inter sect ed by well
MW1 5-59	378160	502351 3	86.261	3.05	83.21	87.0 1	1.52	sand
M W 10 -12	378188	502350 4	85.662	4.28	81.38	86.7 8	1.52	sand
MW1 0-11	378196	502348 6	85.816	4.28	81.53	86.9 6	1.52	sand
MW1 5-51	378205	502350 0	85.933	6.1	79.83	86.7 5	1.52	sand
MW1 0-10	378206	502346 9	85.918	6.73	79.19	86.8 44	1.52	sand

A-3. Hydrostratigraphic data

piezometer #	time (s)	pH	redox potential (mV)
15-51	30	7.32	-46
	60	7.32	-68
	90	7.32	-65
	120	7.32	-70
	150	7.32	-73
	180	7.32	-76
	210	7.32	-78
	240	7.32	-80
	270	7.32	-81
	300	7.32	-82
10--11	30	7.2	103

	60	7.21	98
	90	7.21	95
	120	7.22	92
	150	7.21	89
	180	7.22	87
	210	7.22	84
	240	7.22	83
	270	7.22	81
	300	7.23	80
10--13	30	7.14	99
	60	7.19	44
	90	7.21	11
	120	7.22	8
	150	7.22	-18
	180	7.23	-25
	210	7.23	-29
	240	7.23	-32
	270	7.23	-34
	300	7.23	-37
10--10	30	7.19	55
	60	7.21	18
	90	7.21	-3
	120	7.22	-14
	150	7.22	-21
	180	7.22	-27
	210	7.22	-30
	240	7.22	-32
	270	7.22	-34
	300	7.22	-36
17-70	30	7.17	95
	60	7.17	96
	90	7.18	98
	120	7.18	99

	150	7.18	100
	180	7.18	101
	210	7.18	102
	240	7.18	103
	270	7.18	103
	300	7.18	104
15-59	30	6.6	147
	60	6.59	138
	90	6.59	131
	120	6.58	127
	150	6.58	122
	180	6.57	118
	210	6.56	114
	240	6.56	111
	270	6.56	108
	300	6.56	105
15-52	30	7.42	59
	60	7.43	34
	90	7.43	17
	120	7.44	4
	150	7.44	-4
	180	7.44	-9
	210	7.44	-14
	240	7.43	-17
	270	7.42	-2
	300	7.43	-22
10--12	30	7.01	154
	60	7	154
	90	6.98	152
	120	6.97	150
	150	6.96	147
	180	6.95	144
	210	6.95	141

	240	6.94	139
	270	6.93	137
	300	6.92	134
10--14	30	7	121
	60	6.92	126
	90	6.89	132
	120	6.87	135
	150	6.85	140
	180	6.84	144
	210	6.83	148
	240	6.83	151
	270	6.82	154
	300	6.81	157

A-4. Geochemical field analysis.

alkalinity	H2SO4	
	V added (mL)	Digits
endpoint 1	0.12	96
endpoint 2	0.12125	97
endpoint 3	0.1225	98
eq	0.000194	
CaCO3	388	
alkalinity		
endpoint 1	0.09375	75
endpoint 2	0.095	76
endpoint 3	0.1	80

eq	0.000152	
CaCO3	304	
alkalinity		
endpoint 1	0.06875	55
endpoint 2	0.07	56
endpoint 3	0.07	56
eq	0.000112	
CaCO3	224	
alkalinity		
endpoint 1	0.11875	95
endpoint 2	0.12125	97
endpoint 3	0.12375	99
eq	0.000194	
CaCO3	388	
alkalinity		

endpoint 1	0.0725	58
endpoint 2	0.07375	59
endpoint 3	0.07625	61
eq	0.000118	
CaCO3	236	
alkalinity		
endpoint 1	0.025625	20.5
endpoint 2	0.02875	23
endpoint 3	0.03	24
eq	0.000046	
CaCO3	92	
alkalinity		
endpoint 1	0.04	32
endpoint 2	0.04125	33
endpoint 3	0.0425	34
eq	0.000066	
CaCO3	132	

alkalinity		
endpoint 1	0.0475	38
endpoint 2	0.04875	39
endpoint 3	0.05	40
eq	0.000078	
CaCO3	156	
alkalinity		
endpoint 1	0.11375	91
endpoint 2	0.115	92
endpoint 3	0.11875	95
eq	0.000184	
CaCO3	368	

A-5. Alkalinity analysis.

Piezometer	distance between wells (m)	Distance along the section line(m)	PH	redox potential (mv)	Alkalinity (CaCO ₃ mg/l)	Fe ²⁺ (mg/l)	Fe ³⁺ (mg/l)	Total Fe(mg/l)	So ₄ ²⁻ (mg/l)
MW15-59	0	10	6.56	105	92	0.58	0.39	0.94	15
MW10-12	12	22	6.95	134	156	0.53	0.27	0.3	51
MW10-14	8	30	6.84	157	368	0.04	0	0	0
MW10-11	4	34	7.22	81	304	0.03	0.1	0.33	16
MW17-70	7	41	7.18	104	236	0	0.04	0	31
MW15-51	2	43	7.32	-80	388	1.18	0.13	4.85	20

MW10-10	6	49	7.22	-36	388	0.9	1.21	2.85	43
MW15-52	4	53	7.44	-22	132	0.12	0.64	0.05	18
MW10-13	2	55	7.23	-37	224	1.02	1.24	2.15	42

A-6. Colorimetry Analysis

Location	Piezometer	PH	Redox potential (mv)	Alkalinity (CaCO ₃ mg/l)	Fe ²⁺ (mg/l)	Fe ³⁺ (mg/l)	Total Fe(mg/l)	So ₄ ²⁻ (mg/l)
Inside	MW17-70	7.18	104	236	0	0.04	0	31
	MW10-10	7.22	-36	388	0.9	1.21	2.85	43
	MW10-11	7.22	81	304	0.03	0.1	0.33	16
	MW10-13	7.23	-37	224	1.02	1.24	2.15	42
	MW15-51	7.32	-80	388	1.18	0.13	4.85	20
Outside	MW10-12	6.95	134	156	0.53	0.27	0.3	51
	MW10-14	6.84	157	368	0.04	0	0	0
	MW15-52	7.44	-22	132	0.12	0.64	0.05	18
	MW15-59	6.56	105	92	0.58	0.39	0.94	15

A-7. In vs. Out Plume Analysis

Hydraulic Conductivities		
piezometer	IN	OUT
MW10-10	5.32E-05	4.5052E-05
MW10-11	1.09E-04	7.4608E-05
MW10-12	2.17E-04	1.58E-04
MW15-51	1.74E-04	1.67E-04
MW15-59	2.13982E-05	1.88E-05

A-8. Hydraulic Conductivities Derived from Slug Tests

Piezometer ID	Depth to Water (m)	Water Level (m)
15-62	2.624	3.669
15-56	2.682	3.537
15-55	2.682	3.077
15-57	2.948	4.92
08-01	2.555	4.958
08-02	2.642	5.225
08-05	1.793	3.137
15-58	2.674	3.155
08-03	2.870	5.150
09-9	2.326	3.648
15-59	2.400	3.017
15-57C	2.545	3.071
17-783	2.773	4.245
17-783A	2.681	4.023
10-17	1.146	5.057
15-48C		
10-12		
15-53		
08-4	2.382	4.932
10-14		
10-11		
10-15	2.947	3.426
10-13	2.918	3.439

Sample calculation:

Alkalinity: MW15-59

23 digits of 0.8M H₂SO₄ solution was added to the 25ml of water sample to reach the end point

$$23/800=0.02875 \text{ ml}$$

$$0.02875 \text{ ml} \times 1000 \text{ ml}/25 \text{ ml}=1.15 \text{ ml}=0.00115 \text{ L}$$

$$0.00115 \text{ L} \times 0.8=0.00092 \text{ mol}$$

$$0.00092 \text{ mol} \times 2=0.00184 \text{ mol CaCO}_3/\text{l}$$

$$0.00184 \text{ mol} \times 50000 \text{ mg/L}^*=92 \text{ mg CaCO}_3/\text{L}$$

The Alkalinity of MW15-59 is 92 mg CaCO₃/L

$$\text{Water depth (m)} : 3.669 \text{ m} - 2.624 \text{ m} = 1.045 \text{ m}$$

Hydraulic head: Hydraulic head = Top of the piezometer (m) – water level (m)

(MW08-1) Hydraulic head = 87.25m - 2.555m = 84.695 m

Hydraulic conductivity (K): $K = r^2(\ln(L/R))/2(L)(T_o)$

(MW10-10) $K = (0.0155^2(\ln(1.289/0.0155))/2(1.289)(6.74)$

$K = 6.11E-05 \text{ m/s}$

$T_o \text{ (s)} : 0.37 = 1.1243e^{-0.16 (T_o)}$

$T_o = 6.818 \text{ s}$

Specific discharge (q) = $-K (dh/dL)$

(MW15-59) $q = -1.87635E-05 \text{ m/s (0.100m/30m)}$

$q = -6.255E-08 \text{ (m}^3/\text{m}^2/\text{s)}$